

SEMESTER II EXAMINATION 2025-2026

GENAI – Generative AI

May 2026

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

1. This examination paper contains **eight (8)** questions and comprises **two (2)** printed pages.
 2. Answer all questions. The marks for each question are indicated at the beginning of each question.
 3. Answer concisely but precisely. Use diagrams or examples when useful. For project-related questions, support your answer with observations from your own experiments.
 4. This **IS NOT** an **OPEN BOOK** exam.
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Question 1. (10 marks)

Harold Cohen's AARON and Evolutionary Art systems represent two different philosophies for building generative systems. Compare these classical approaches in terms of the source of "creativity", their capacity for surprise, and the role of the human designer.

Question 2. (12.5 marks)

A Variational Autoencoder (VAE) is trained by optimizing a loss with two main terms. Explain the role of each term and what happens to the generated outputs when one dominates the other.

Question 3. (12.5 marks)

Explain the adversarial training dynamic of GANs and describe mode collapse as a failure mode, including how you would detect it.

$$L_{GAN} = \log(D(x)) + \log(1 - D(G(z)))$$

Question 4. (12.5 marks)

Describe the forward and reverse processes in DDPMs at a conceptual level. Then explain classifier-free guidance both in terms of mechanism and impact in image generation.

Question 5. (12.5 marks)

Autoregressive language models generate text one token at a time, and the sampling strategy strongly affects the output. Compare temperature scaling, top-k, and nucleus (top-p) sampling as three ways of controlling the balance between diversity and coherence.

Question 6. (12.5 marks)

You trained VAE, DCGAN, and DDPM on ArtBench-10. Each model family has well-known theoretical properties that predict the kind of outputs it tends to produce. Discuss whether your experimental results confirmed or contradicted these theoretical expectations, and explain why. If one of your models performed surprisingly well or poorly on specific art styles, propose an explanation rooted in how that model family works.

Question 7. (12.5 marks)

The project used FID and KID as the primary evaluation metrics. Critically evaluate this evaluation protocol: what aspects of generation quality do FID and KID capture well, what do they miss, and how could the protocol be misleading? If you could add one additional evaluation method to better understand the differences between your three models, what would it be and why?

Question 8. (15 marks)

Suppose you were now asked to repeat the project at 256×256 resolution instead of 32×32 . For each type of model you explored, discuss what would break or need to change, and which model family would benefit the most from the increased resolution. Would you still use the same three architectures, or would you replace any of them with a different approach? Justify your answer based on what you learned from the project.

END OF PAPER